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Batch: CS2(C21)

CODETANTRA LAB PRACTICAL 5

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5.1.1. Stacked Plot

25/04

Create a stacked area plot to visualize the temperature variations for three different cities (City A, City B, and City C) across the months of the year. The temperature data is provided for each city in the editor.

Your task is to:

- Create a stacked area plot using the data.
- Label the x-axis as "Month", the y-axis as "Temperature", and provide the title "Temperature Variation" for the plot.
- Display the plot showing the temperature variation for each city throughout the months of the year.

Sample Test Cases

stackedpl...

1 import matplotlib.pyplot as plt
2 import pandas as pd
3
4 # Data for Months and Temperature for three cities
5 data = {
6 'Month': ['January', 'February', 'March', 'April', 'May', 'June',
7 'July', 'August', 'September', 'October', 'November', 'December'],
8 'City_A_Temperature': [5, 7, 10, 13, 17, 20, 22, 21, 18, 12, 8, 6],
9 'City_B_Temperature': [2, 3, 5, 6, 10, 14, 16, 17, 12, 9, 5, 3],
10 'City_C_Temperature': [3, 4, 6, 8, 9, 12, 15, 14, 10, 7, 4, 2]
11 }
12
13 # Write your code...
14 df=pd.DataFrame(data)
plt.stackplot(df['Month'],df['City_A_Temperature'],df['City_B_Temperature'])

Average time 0.236 s
Maximum time 0.236 s
1 out of 1 shown test case(s) passed

Test case 1 0.236 ms
Expected output
Actual output

Terminal Test cases

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5.2.1. Titanic Dataset

01/27

Write a Python program to analyze and visualize data from the Titanic dataset based on the following instructions:

Dataset Information:
The dataset is stored in a CSV file named `titanic.csv` and has been loaded using the `pandas` library. It contains the following columns:

- `Pclass`: Passenger class (1 = First, 2 = Second, 3 = Third).
- `Gender`: Gender of the passenger (male/female).
- `Age`: Age of the passenger.
- `Survived`: Survival status (0 = Did not survive, 1 = Survived).
- `Fare`: Ticket fare paid by the passenger.

Visualization:
To represent these trends, you will create 5 visualizations using Matplotlib. The visualizations should be arranged in a 3x2 grid (3 rows and 2 columns).

Visualization Details:
Write the code to create a series of visualizations as follows:

Bar Plot (Pclass Distribution):

- Create a bar plot to show the distribution of passengers across the different passenger classes (`Pclass`).
- Use the color `skyblue` for the bars.
- Title the plot as "Passenger Class Distribution".
- Label the x-axis as "Pclass" and the y-axis as "Count".

Pie Chart (Gender Distribution):

- Create a pie chart to display the distribution of male and female passengers.
- Use `skyblue` for males and `red` for females.

titanicDat...

1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset from the CSV file
5 df = pd.read_csv('titanic.csv')
6
7 # Set up the figure for 5 subplots
8 fig, axes = plt.subplots(3, 2, figsize=(12, 12))
9
10 # write the code..
11 count=df['Pclass'].value_counts().sort_index()
12 count_gen=df['Gender'].value_counts()
13 count_s=df['Survived'].value_counts().sort_index()
14
15 # write the code..

Average time 1.070 s
Maximum time 1.070 s
1 out of 1 shown test case(s) passed

Test case 1 1.070 ms
Expected output
Actual output

Terminal Test cases

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5.2.2. Histogram of passenger information of Titanic

19:06

Write a Python code to plot a histogram for the distribution of the 'Age' column from the Titanic dataset. The histogram should display the frequency of different age ranges with the following specifications:

- Use 30 bins for the histogram.
- Set the edge color of the bars to black (k).
- Label the x-axis as 'Age' and the y-axis as 'Frequency'.
- Add the title "Age Distribution" to the histogram.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Histogram...

1import pandas as pd
2import matplotlib.pyplot as plt
3
4# Load the Titanic dataset
5data = pd.read_csv('Titanic-Dataset.csv')
6
7# Data Cleaning
8data['Age'].fillna(data['Age'].median(), inplace=True)
9data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10data.drop('Cabin', axis=1, inplace=True)
11
12# Convert categorical features to numeric
13data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15

Average time0.266 sMaximum time0.266 s1 out of 1 shown test case(s) passed

Test case 10.00 msDebugTest cases

Expected outputActual output

Age DistributionAge Distribution

TerminalTest cases

< PrevResetSubmitNext>

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5.2.3. Bar plot of survival rate of passengers

09:00

Write a Python code to plot a bar chart that shows the count of passengers who survived and did not survive in the Titanic dataset. The chart should display the following specifications:

- Use the 'Survived' column to show the count of survivors (0 = Did not survive, 1 = Survived).
- Set the chart type to 'bar'.
- Add the title "Survival Count" to the chart.
- Label the x-axis as 'Survived' and the y-axis as 'Count'.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

BarPlotOf...

1import pandas as pd
2import matplotlib.pyplot as plt
3
4# Load the Titanic dataset
5data = pd.read_csv('Titanic-Dataset.csv')
6
7# Data Cleaning
8data['Age'].fillna(data['Age'].median(), inplace=True)
9data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10data.drop('Cabin', axis=1, inplace=True)
11
12# Convert categorical features to numeric
13data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15

Average time0.306 sMaximum time0.306 s1 out of 1 shown test case(s) passed

Test case 10.00 msDebugTest cases

Expected outputActual output

Survival CountSurvival Count

TerminalTest cases

< PrevResetSubmitNext>

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5.2.4. Bar Plot for Survival by Gender

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by gender, in the Titanic dataset. The chart should display the following specifications:

- Group the data by the 'Sex' column, then use the `value_counts()` function to count the occurrences of survivors (0 = Did not survive, 1 = Survived) for each gender.
- Use a **stacked bar chart** to display the survival counts.
- Add the title "Survival by Gender" to the chart.
- Label the x-axis as 'Gender' and the y-axis as 'Count'.
- The legend should indicate 'Not Survived' and 'Survived'.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171	7.25	S	
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599	71.2833	C85,C	
3	1	3	Heikinen, Miss. Laina	female	26	0	0	STON/O2. 3101282	7.925	S	
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803	53.1	C123,S	
5	0	3	Allen, Mr. William Henry	male	35	0	0	373450	8.05	S	
6	0	3	Moran, Mr. James	male	0	0	0	330877	8.4583	Q	
7	0	1	McCarthy, Mr. Timothy J	male	54	0	0	17463	51.8625	E46,S	
8	0	3	Palsson, Master. Gosta Leonard	male	2	3	1	349909	21.075	S	
9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27	0	2	347742	11.1333	S	

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1, 0, 3, "Braund, Mr. Owen Harris", male, 22, 1, 0, A/5 21171, 7.25, S
2, 1, 1, "Cumings, Mrs. John Bradley (Florence Briggs Thayer)", female, 38, 1, 0, PC 17599, 71.2833, C85, C
3, 1, 3, "Heikinen, Miss. Laina", female, 26, 0, 0, STON/O2. 3101282, 7.925, S
4, 1, 1, "Futrelle, Mrs. Jacques Heath (Lily May Peel)", female, 35, 1, 0, 113803, 53.1, C123, S
5, 0, 3, "Allen, Mr. William Henry", male, 35, 0, 0, 373450, 8.05, S
6, 0, 3, "Moran, Mr. James", male, 0, 0, 330877, 8.4583, Q
7, 0, 1, "McCarthy, Mr. Timothy J", male, 54, 0, 0, 17463, 51.8625, E46, S
8, 0, 3, "Palsson, Master. Gosta Leonard", male, 2, 3, 1, 349909, 21.075, S
9, 1, 3, "Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)", female, 27, 0, 2, 347742, 11.1333, S
```

BarPlotOf...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
```

Average time0.397 s397.00 msMaximum time0.397 s397.00 ms1 out of 1 shown test case(s) passed

Test case 1397 msDebug

Expected outputActual output

Survival by GenderSurvival by Gender

TerminalTest cases

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5.2.5. Bar Plot for Survival by Pclass

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by passenger class (Pclass), in the Titanic dataset. The chart should display the following specifications:

- Group the data by the **Pclass** column and count the number of survivors (0 = Did not survive, 1 = Survived) for each class using `value_counts()`.
- Use a **stacked bar chart** to display the survival counts.
- Add the title "Survival by Pclass" to the chart.
- Label the x-axis as 'Pclass' and the y-axis as 'Count'.
- The legend should indicate 'Not Survived' and 'Survived'.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171	7.25	S	
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599	71.2833	C85,C	
3	1	3	Heikinen, Miss. Laina	female	26	0	0	STON/O2. 3101282	7.925	S	
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803	53.1	C123,S	
5	0	3	Allen, Mr. William Henry	male	35	0	0	373450	8.05	S	
6	0	3	Moran, Mr. James	male	0	0	0	330877	8.4583	Q	
7	0	1	McCarthy, Mr. Timothy J	male	54	0	0	17463	51.8625	E46,S	
8	0	3	Palsson, Master. Gosta Leonard	male	2	3	1	349909	21.075	S	
9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27	0	2	347742	11.1333	S	

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1, 0, 3, "Braund, Mr. Owen Harris", male, 22, 1, 0, A/5 21171, 7.25, S
2, 1, 1, "Cumings, Mrs. John Bradley (Florence Briggs Thayer)", female, 38, 1, 0, PC 17599, 71.2833, C85, C
3, 1, 3, "Heikinen, Miss. Laina", female, 26, 0, 0, STON/O2. 3101282, 7.925, S
4, 1, 1, "Futrelle, Mrs. Jacques Heath (Lily May Peel)", female, 35, 1, 0, 113803, 53.1, C123, S
5, 0, 3, "Allen, Mr. William Henry", male, 35, 0, 0, 373450, 8.05, S
6, 0, 3, "Moran, Mr. James", male, 0, 0, 330877, 8.4583, Q
7, 0, 1, "McCarthy, Mr. Timothy J", male, 54, 0, 0, 17463, 51.8625, E46, S
8, 0, 3, "Palsson, Master. Gosta Leonard", male, 2, 3, 1, 349909, 21.075, S
9, 1, 3, "Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)", female, 27, 0, 2, 347742, 11.1333, S
```

BarPlotOf...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
```

Average time0.412 s412.00 msMaximum time0.412 s412.00 ms1 out of 1 shown test case(s) passed

Test case 1412 msDebug

Expected outputActual output

Survival by PclassSurvival by Pclass

TerminalTest cases

< PrevResetSubmitNext >

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5.2.6. Bar Plot for Survival by Embarked

Write a Python code to plot a stacked bar chart showing the survival count for passengers based on their embarkation location in the Titanic dataset.
The chart should display the following specifications:
1. Use the **Embarked** column to determine the embarkation location. After converting this column into dummy variables (using `pd.get_dummies()`), plot the survival count based on the **Embarked_Q** column (representing passengers who embarked from Queenstown) in relation to survival.
2. Set the chart type to 'bar' and make it stacked.
3. Add the title **"Survival by Embarked"** to the chart.
4. Label the x-axis as **'Embarked'** and the y-axis as **'Count'**.
5. Include a legend to distinguish between survivors and non-survivors (label the legend as **'Survived'** and **'Not Survived'**).

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171,7.25,,S			
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599,71.2833,C85,C			
3	1	3	Heikkinen, Miss. Laina	female	26	0	0	STON/O2. 3101282,7.925,,S			
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803,53.1,C123,S			
5	0	3	Allen, Mr. William Henry	male	35	0	0	373450,8.05,,S			

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

BarPlotOf...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
```

Average time: 0.412 s (412.00 ms) Maximum time: 0.412 s (412.00 ms) 1 out of 1 shown test case(s) passed

Test case 1 412 ms Debug

Expected output: Survival by Embarked (Stacked bar chart showing survival counts for Embarked categories: Q, S, C. Legend: Not Survived (blue), Survived (orange).)

Actual output: Survival by Embarked (Stacked bar chart showing survival counts for Embarked categories: Q, S, C. Legend: Not Survived (blue), Survived (orange).)

Terminal Test Cases

< Prev Reset Submit Next >

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5.2.7. Box plot for Age Distribution

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset across different passenger classes. The boxplot should display the following specifications:
1. Use the **Pclass** column to group the data for the boxplot.
2. Set the title of the plot to **"Age by Pclass"**.
3. Remove the default subtitle with `plt.suptitle("")`.
4. Label the x-axis as **'Pclass'** and the y-axis as **'Age'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171,7.25,,S			
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599,71.2833,C85,C			
3	1	3	Heikkinen, Miss. Laina	female	26	0	0	STON/O2. 3101282,7.925,,S			
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803,53.1,C123,S			
5	0	3	Allen, Mr. William Henry	male	35	0	0	373450,8.05,,S			
6	0	3	Moran, Mr. James	male	0	0	0	330877,8.4583,,Q			
7	0	1	McCarthy, Mr. Timothy J	male	54	0	0	17463,51.8625,E46,S			
8	0	3	Palsson, Master. Gosta Leonard	male	2	3	1	349909,21.075,,S			
9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27	0	2	347742,11.1333,,S			
10	1	2	Nasser, Mrs. Nicholas (Adele Achem)	female	14	1	0	237736,30.0708,,C			

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

BoxPlotF...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
```

Average time: 0.371 s (371.00 ms) Maximum time: 0.371 s (371.00 ms) 1 out of 1 shown test case(s) passed

Test case 1 371 ms Debug

Expected output: Age by Pclass (Boxplot showing age distribution for Pclass categories: 1, 2, 3. Y-axis: Age. X-axis: Pclass.)

Actual output: Age by Pclass (Boxplot showing age distribution for Pclass categories: 1, 2, 3. Y-axis: Age. X-axis: Pclass.)

Terminal Test Cases

< Prev Reset Submit Next >

5.2.8. Box Plot for Age by Survived

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset based on whether passengers survived or not. The boxplot should display the following specifications:

1. Use the **Survived** column to group the data for the boxplot (0 = Did not survive, 1 = Survived).
2. Set the title of the plot to **"Age by Survival"**.
3. Remove the default subtitle with `plt.suptitle("")`.
4. Label the x-axis as **'Survived'** and the y-axis as **'Age'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171	7.25	S	
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599	71.2833	C85	C
3	1	3	Heikkinen, Miss. Laina	female	26	0	0	STON/O2. 3101282	7.925	S	
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803	53.1	C123	S
5	0	3	Allen, Mr. William Henry	male	35	0	0	373450	8.05	S	
6	0	3	Moran, Mr. James	male	0	0	0	330877	8.4583	Q	
7	0	1	McCarthy, Mr. Timothy J	male	54	0	0	17463	51.8625	E46	S
8	0	3	Palsson, Master. Gosta Leonard	male	2	3	1	349909	21.075	S	
9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27	0	2	347742	11.1333	S	
10	1	2	Nasser, Mrs. Nicholas (Adele Achem)	female	14	1	0	237736	30.0708	C	

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

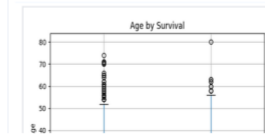
BoxPlotF...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
```

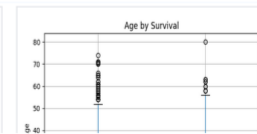
Average time 0.351 s 351.00 ms Maximum time 0.351 s 351.00 ms 1 out of 1 shown test case(s) passed

Test case 1 351 ms

Expected output



Actual output



Terminal Test cases

< Prev Reset Submit Next >

5.2.9. Box Plot for Fare by Pclass

Write a Python code to plot a boxplot that shows the distribution of the 'Fare' column from the Titanic dataset based on the passenger class (Pclass). The boxplot should display the following specifications:

1. Use the **Pclass** column to group the data for the boxplot.
2. Set the title of the plot to **"Fare by Pclass"**.
3. Remove the default subtitle with `plt.suptitle("")`.
4. Label the x-axis as **'Pclass'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171	7.25	S	
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599	71.2833	C85	C
3	1	3	Heikkinen, Miss. Laina	female	26	0	0	STON/O2. 3101282	7.925	S	
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803	53.1	C123	S
5	0	3	Allen, Mr. William Henry	male	35	0	0	373450	8.05	S	
6	0	3	Moran, Mr. James	male	0	0	0	330877	8.4583	Q	
7	0	1	McCarthy, Mr. Timothy J	male	54	0	0	17463	51.8625	E46	S
8	0	3	Palsson, Master. Gosta Leonard	male	2	3	1	349909	21.075	S	
9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27	0	2	347742	11.1333	S	
10	1	2	Nasser, Mrs. Nicholas (Adele Achem)	female	14	1	0	237736	30.0708	C	

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

BoxPlotF...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
```

Average time 0.365 s 365.00 ms Maximum time 0.365 s 365.00 ms 1 out of 1 shown test case(s) passed

Test case 1 365 ms

Expected output



Actual output



Terminal Test cases

< Prev Reset Submit Next >

5.2.10. Scatter Plot for Age vs. Fare

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Set the title of the plot to **"Age vs. Fare"**.
3. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,0,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

Note: Refer to the visible test case for better reference.

AgeFareS...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
```

Average time 0.247 s Maximum time 0.247 s 1 out of 1 shown test case(s) passed



Terminal Test cases

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5.2.11. Scatter Plot for Age vs. Fare by Survived

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset, with points color-coded by survival status. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Color the points based on the **Survived** column: **Red** for passengers who did not survive (**Survived = 0**). **Blue** for passengers who survived (**Survived = 1**).
3. Set the title of the plot to **"Age vs. Fare by Survival"**.
4. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

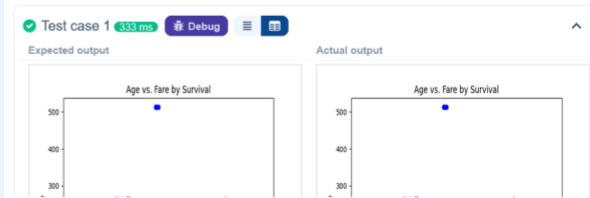
Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,0,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

AgeFareS...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
```

Average time 0.333 s Maximum time 0.333 s 1 out of 1 shown test case(s) passed



Terminal Test cases

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